Name: _____

Class: _____

Date: _____ **Mr. Rawson • GCSE Physics**

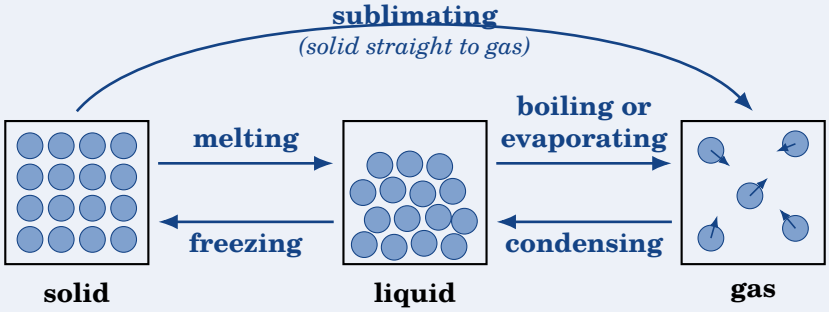
3.3 Changes of State

Five changes, the same particles throughout. Name them, count them, and use density across them.

1 · Naming the changes five arrows between three states

Key concept — the five changes of state

Heating a substance can change its state; cooling it changes the state back. Each arrow below is one change, and each change has a name. Arrows **above** the boxes happen when the substance is **heated**; arrows **below** happen when it is **cooled**.



Boiling and evaporating both turn a liquid into a gas. **Boiling** happens throughout the liquid, at one fixed temperature: the **boiling point**. **Evaporating** happens only at the surface, and at any temperature below the boiling point. A puddle dries up by evaporating; a kettle boils.

The same changes are also written as nouns: *melting, freezing, vaporisation, condensation, sublimation*. Either form is fine in an answer.

1) **Complete** the table. Each row is one change of state. **[5]**

From	To	Name of the change
solid	liquid	*melting*
liquid	solid	freezing
liquid	gas	*boiling (or evaporating)*
gas	*liquid*	condensing
solid	*gas*	sublimating

Check: cover the diagram and read your table aloud as five sentences, *"a solid melts into a liquid"*. Each one follows the direction of its arrow.

2 · Mass is conserved — the same particles, rearranged

Key concept — a change of state keeps every particle

When a substance changes state, its particles move apart or closer together and are rearranged, but they are **the same particles**: none are added and none are taken away. The number of particles stays the same, so the **mass stays the same**: **conservation of mass**.

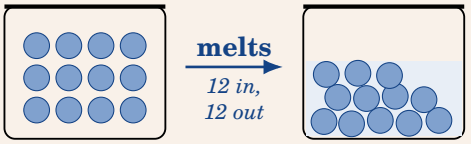
Worked example — a sealed bag of ice

A sealed freezer bag holds 250 g of ice. It is left on a bench until all the ice has melted. What does the balance read now?

1. Count The bag is sealed: no particles leave and none get in.

2. Compare The ice particles are rearranged into water. Same particles, same number.

3. Answer Mass is conserved, so the balance reads **250 g**.



ice, sealed
balance: **250 g**

water, same bag
balance: **250 g**

2) **Explain** this observation. An open beaker holding 200 g of water is heated until it boils, and left boiling for ten minutes. The balance now reads 180 g. Say where the missing 20 g is, and whether mass has been conserved. **[3]**

The 20 g of water has boiled and left the beaker as a gas (water vapour) which has spread into the room. The particles still exist, so mass is conserved overall: 180 g in the beaker plus 20 g of water vapour in the air is still 200 g. The beaker is open, so the balance only measures what is left in it.

Check: your answer to question 2 says where the *particles* are now.

3 · Physical or chemical? — reverse it, and see what comes back

Key concept — two kinds of change

A physical change

No new substance is made: the particles are the same, only their arrangement, spacing and movement change. **Reverse the change** and the material **recovers its original properties**: melt ice, freeze the water, and you have ice again.

Every change of state is a physical change.

A chemical change

A new substance is made: the particles are joined up differently, so it has its own properties. Cooling or heating it back does not return the original: the ash from burnt paper stays ash.

Burning, rusting and baking are chemical changes.

Worked example — a candle

Near the flame the wax **melts** and runs down the side. At the wick the wax **burns**.

1. New substance? Melting: no, it is still wax. Burning: yes, it becomes gases (carbon dioxide and water vapour) and soot.

2. Reverse it Cool the melted wax: solid wax, same properties. Cool the gases: still gases.

3. Decide Melting the wax is **physical**. Burning the wax is **chemical**.

3) Sort each change as physical or chemical, and give the reason you used. **[5]**

Change	Physical or chemical?	How you can tell
Chocolate melting in your hand	*physical*	*still chocolate; cool it and it sets again*
Paper burning	*chemical*	*new substances (ash, gases); cooling does not give paper back*
Steam condensing on a cold window	*physical*	*still water; warm it and it evaporates again*
An iron nail rusting	*chemical*	*rust is a new substance with different properties*
Cake batter baking into a cake	*chemical*	*the cake is a new substance; cooling it does not give batter*

Check: every row marked *physical* names what comes back when the change is reversed.

4 · Density across a change of state — same mass, new volume

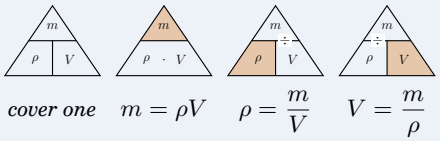
Key concept — carry the mass across

Recall the density equation: it is not on your equation sheet.

$$\rho = \frac{m}{V} \qquad \text{density} = \frac{\text{mass}}{\text{volume}}$$

ρ in kg/m^3, m in kg, V in m^3.

When the state changes, **the mass is conserved**: m is the same number before and after. The **volume changes**, and so does the density.



Useful densities: water 1000 kg/m^3, ice 920 kg/m^3, steam 0.60 kg/m^3.

Worked example — a bottle of water in the freezer

A sealed bottle holds 0.90 kg of water. It is put in a freezer and all the water turns to ice. Calculate the volume of the water, then the volume of the ice.

1. Carry the mass — Mass is conserved: $m = 0.90$ kg before *and* after.

2. Rearrange — $V = m/\rho$

3. Water, then ice — $V = 0.90/1000 = \mathbf{0.00090}$ m^3 $\qquad$ $V = 0.90/920 = \mathbf{0.00098}$ m^3

4. Compare — Same mass, larger volume: the ice takes up about 9% more room. A full, sealed bottle has nowhere for that extra volume to go, which is why it can crack in a freezer.

4) Calculate the volume of steam produced when 0.50 kg of water boils away completely. Density of steam = 0.60 kg/m^3. Show the equation, the substitution and the answer with its unit. **[3]**

Mass is conserved, so the steam has $m = 0.50$ kg. $V = m/\rho = 0.50/0.60 = \mathbf{0.83}$ m^3 (0.833…). As water it filled only 0.00050 m^3: the same particles now fill about 1700 times the volume.

5) **Calculate** the mass of a block of ice of volume $0.0050\,\text{m}^3$ (density of ice $= 920\,\text{kg/m}^3$), and then the volume of the water formed when it melts in a sealed container (density of water $= 1000\,\text{kg/m}^3$). **[5]**

$m = \rho V = 920 \times 0.0050 =$ **4.6 kg**. *Mass is conserved, so the water has* $m = 4.6\,kg$*:* $V = m/\rho = 4.6/1000 =$ **0.0046** m^3. *Smaller than the ice: the particles have moved closer together.*

> **Check:** the mass line reads the same number "before" and "after"; the volume line is the one that changes.

5 · Putting it together a new situation, done unaided

6) A 500 g block of solid carbon dioxide (dry ice) is placed in an open dish on a balance in a warm room. Over the next hour the block shrinks, no liquid ever appears in the dish, and the balance reading falls steadily to the mass of the empty dish.

(a) Name the change of state that has taken place.

Sublimating (sublimation): solid straight to gas.

[1]

(b) Explain what the falling balance reading shows, and why mass has still been conserved.

The carbon dioxide particles have left the dish as a gas and spread into the room, so the dish holds less and less. The particles still exist, 500 g of carbon dioxide gas is now in the air, so the total mass is unchanged. The balance only measures what is in the dish.

[3]

(c) State whether this is a physical or a chemical change, and give the reason.

Physical. No new substance is made: it is carbon dioxide before and after. Cool the gas enough and it becomes solid carbon dioxide again, with its original properties.

[2]